Negar Neda

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Education

New York University, Tandon School of Engineering *Sep. 2021 -2026*
PhD in Electrical and Computer Engineering, Advisor: ***Brandon Reagen***
Research Focus: Hardware Acceleration, High Performance Systems, and Performance Modeling.

University of Tehran (UT), Tehran, Iran *Sep. 2018 - Feb. 2021*
M.Sc. in Computer Architecture, GPA: 3.63/4
Thesis title: FPGA-based Multi-precision Accelerator for Deep Neural Networks

Amirkabir University of Technology (AUT), Tehran, Iran *Sep. 2014 - Sep. 2018*
B.Sc. in Computer Architecture, GPA: 3.62/4

Current Research

I design **high-performance hardware accelerators** for compute-intensive workloads, focusing on Fully Homomorphic Encryption (**FHE**) and **LLMs**. My work involves **hardware-algorithm co-design**, **architecture modeling**, and **performance analysis** for **large-scale linear algebra kernels** used in **machine learning** and **cryptographic** algorithms. I have experience optimizing matrix multiplication for encrypted data, dataflow scheduling, memory hierarchies, and parallel execution across accelerators and chiplet-based architectures.

Publications

· "C2ME: Characterizing Ciphertext Matrix Multiplication Encodings for FHE Inference", International Symposium on Workload Characterization (IISWC), **N. Neda**, A. Mahajan, B. Reagen, 2026

· "Network and Compiler Optimizations for Efficient Linear Algebra Kernels in Private Transformer Inference (Invited Paper)", International Conference on Computer-Aided Design (ICCAD), **N. Neda**, K. Garimella, A. Ebel, N. Kumar Jha, B. Reagen, 2025

· "CiFlow: Dataflow Analysis and Optimization of Key Switching for Homomorphic Encryption", IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), **N. Neda**, A. Ebel, B. Reynwar, B. Reagen, 2024.

· "Quantifying the Overheads of Modular Multiplication", IEEE International Symposium on Low Power Electronics and Design (ISLPED), D. Soni, M. Nabeel, **N. Neda**, et al., 2023.

· "Towards fast and scalable private inference", Proceedings of the 20th ACM International Conference on Computing Frontiers, J. Mo, K. Garimella, **N. Neda**, A. Ebel, B. Reagen, 2023.

· "RPU: The Ring Processing Unit", IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), **N. Neda**, D. Soni, et al., 2023.

· "Multi-Precision Deep Neural Network Acceleration on FPGAs", Asia and South Pacific Design Automation Conference (ASP-DAC), **N. Neda**, et al., 2022.

Research Experience

Graduate Research Assistant, *New York University* 2021 - Present
· Designed high-performance hardware accelerators for compute-intensive workloads.
· Built performance & architecture models in Python and C++ to analyze system trade-offs.
· Implemented and optimized high-performance linear algebra and FFT/NTT on encrypted data.
· Optimized dataflows and memory hierarchies to reduce off-chip bandwidth and improve utilization.

Research Assistant, *University of Tehran* 2018 - 2021

· Implemented an FPGA-based Multi-precision accelerator for DDNs.

Teaching Experience

· Computing Systems Architecture · Computer Aided Digital System Design · Logic Circuit Laboratory

Selected Projects

Accelerating Encrypted LLM Inference with FHE *2025 - ongoing*

· Exploring hardware-algorithm co-design, GPU kernels, and packing-aware optimizations to accelerate encrypted LLM decoder inference, focusing on attention and nonlinear functions.

Characterizing Matrix Multiplication Encodings for FHE Inference *2025 – 2026*

· Built a unified Lattigo framework to compare ciphertext-ciphertext matrix multiplication encodings and characterize their runtime, memory, energy, and operation-count tradeoffs.

Chiplet-Based Accelerator for TFHE *2024 – ongoing*

· Designed a **chiplet-based** accelerator for **TFHE** programmable bootstrapping, achieving up to **2×** speedup and throughput without HBM, with a compact $\mathbf{13\,mm^2}$ chiplet area.

Optimized Key-Switching Dataflow for Efficient Homomorphic Encryption *2023*

· Improved data reuse by strategically scheduling key-switching operations in HE, achieving **4×** speedup and **12×** SRAM savings by minimizing off-chip data movement.

Vector-based ISA & Microarchitecture for NTT Acceleration *2022*

· Developed a **vector ISA** and custom hardware for accelerating RLWE-based workloads, achieving **1485×** speedup for **NTT** over CPU using modular arithmetic support and parallel processing.

5-Stage MIPS Processor with Pipelining *2021*

· Built a pipelined **MIPS processor** in C++ with a 2-bit branch predictor and a direct/fully associative cache implementation.

Multi-Precision Accelerator for DNN on FPGA *2021*

· Designed a dynamic bit-width multiplier for **DNNs** on **FPGA**, enabling 3–15× throughput by dynamically adjusting bit-width to match precision needs.

HONOR & AWARDS

· Selected for a Silicon Engineering Internship at Google — Summer 2026
· Awarded a Student Travel Grant to present our work at ISPASS — 2023 & 2024
· Earned **Future Leader Fellowship** at NYU Tandon. — 2022-2024
· **Ranked Top 3 in terms of GPA**, among Computer Architecture Students in AUT — 2019

Relevant Courses

· Computing Systems Architecture · Intro to Cryptography · Hardware Security · Machine Learning
· Neural Networks · Multicore Embedded Systems · VLSI Systems Design

Technical skills

Programming	Python, C/C++, CUDA, Rust, System/Verilog, VHDL
Tools	Git, Linux, NumPy/Pandas, Visual Studio, Vivado Design Suite, MATLAB
Concepts	Algorithms, Data Structures, Systems Performance, Parallel Computing

Talks

· "CiFlow: Dataflow Analysis & Optimization of Key Switching for HE", ISPASS — 2024
· "RPU: The Ring Processing Unit", ISPASS — 2023
· "Summary of RPU: The Ring Processing Unit ", Princeton Computer Architecture Day — 2023
· "A Novel Vector ISA for Accelerating Homomorphic Encryption", TECHCON — 2022